

TASK1 DOCUMENTATION

Project Title:

Data Cleaning and Preprocessing of Dataset for Data Analytics

Abstract

Data cleaning is a crucial step in the data analytics process, as the quality of data directly affects the accuracy of analysis and decision-making. The objective of this project is to clean and preprocess the "Dataset for Data Analytics" by identifying and handling missing values, removing duplicate records, correcting inconsistencies, and ensuring data quality. The cleaned dataset can be effectively used for data analysis, visualization, and predictive modeling.

1. Introduction

Data collected from various sources often contains errors, missing values, duplicate records, and inconsistencies. These issues can lead to inaccurate analytical results and poor decision-making. Data cleaning is the process of detecting and correcting such issues to improve the quality and reliability of data.

This project focuses on cleaning and preprocessing the "Dataset for Data Analytics" using Python and Pandas. Various techniques were applied to identify and handle data quality issues and prepare the dataset for further analysis.

2. Objectives

- To identify and handle missing values in the dataset.
- To detect and remove duplicate records.
- To standardize data formats and data types.
- To improve the quality and reliability of the dataset.
- To prepare the dataset for further analysis and visualization.

3. Problem Statement

Raw datasets frequently contain missing values, duplicate records, inconsistent formatting, and incorrect entries. Such issues negatively impact data analysis and machine learning models. The objective of this project is to clean and preprocess the dataset to improve its quality and usability.

4. Dataset Description

Dataset Name: Dataset for Data Analytics

The dataset contains records that require cleaning and preprocessing before analysis. The dataset was examined to identify missing values, duplicate records, formatting inconsistencies, and other data quality issues.

Dataset Information

- Dataset Name: Dataset for Data Analytics
- File Format: Excel (.xlsx)
- Domain: Data Analytics
- Purpose: Data Cleaning and Preprocessing

5. Tools and Technologies Used

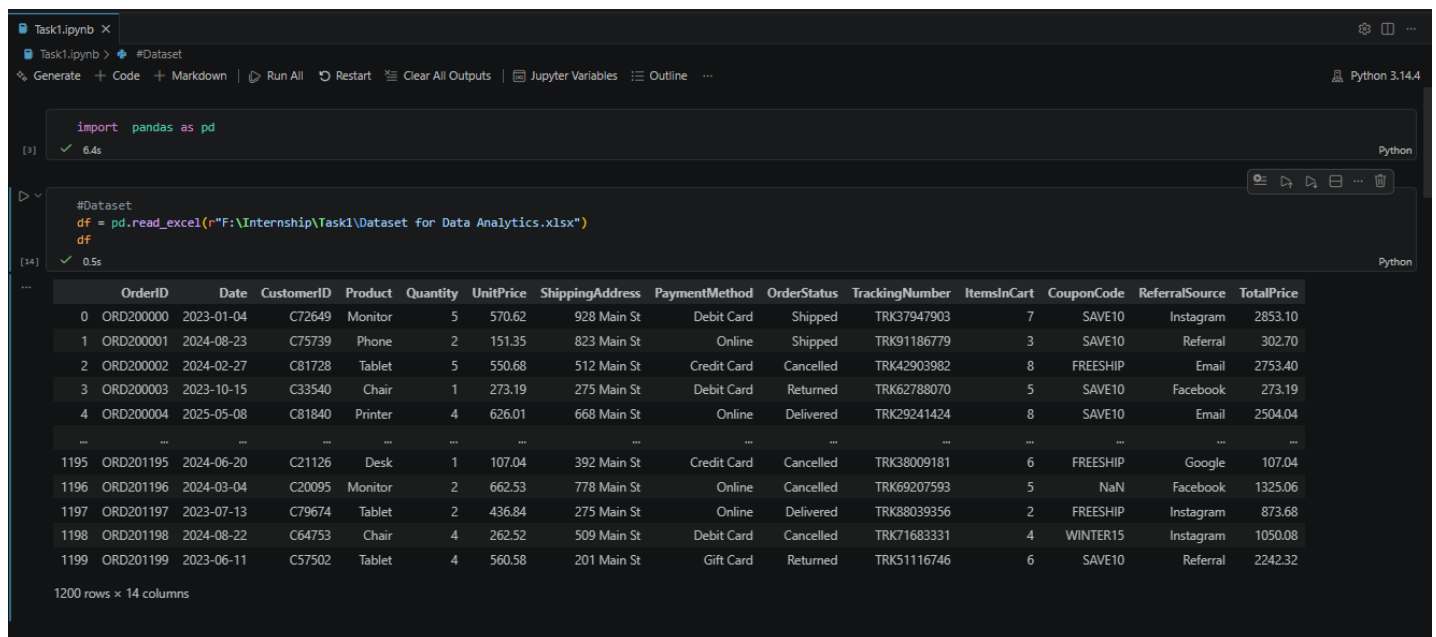
- Jupyter Notebook
- Python
- Pandas
- Microsoft Excel

6. Methodology

The project was completed through the following steps:

Step 1: Data Loading

The dataset was imported into Jupyter Notebook using the Pandas library.



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
import pandas as pd

#Dataset
df = pd.read_excel(r"F:\Internship\Task1\Dataset for Data Analytics.xlsx")
df
```

The output displays a preview of the DataFrame with 14 columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice. The first few rows are shown, followed by an ellipsis indicating more rows, and then rows 1195 through 1199. At the bottom, it states "1200 rows x 14 columns".

Step 2: Data Inspection

The structure of the dataset was examined using:

- Duplicates analysis
- Dataformat check
- Missing Value Analysis

```
#Shape of dataset
print("Shape of dataset:",df.shape)
#Dataset information
df.info()

[18] ✓ 2.5s Python
```

```
-- Shape of dataset: (1200, 14)
-- <class 'pandas.DataFrame'>
-- RangeIndex: 1200 entries, 0 to 1199
-- Data columns (total 14 columns):
-- #   column                Non-Null Count  Dtype
-- ---  ---
-- 0   OrderID                1200 non-null   str
-- 1   Date                  1200 non-null   datetime64[us]
-- 2   CustomerID            1200 non-null   str
-- 3   Product               1200 non-null   str
-- 4   Quantity              1200 non-null   int64
-- 5   UnitPrice             1200 non-null   float64
-- 6   ShippingAddress        1200 non-null   str
-- 7   PaymentMethod         1200 non-null   str
-- 8   OrderStatus           1200 non-null   str
-- 9   TrackingNumber        1200 non-null   str
-- 10  ItemsInCart           1200 non-null   int64
-- 11  CouponCode            1200 non-null   str
-- 12  ReferralSource        1200 non-null   str
-- 13  TotalPrice            1200 non-null   float64
-- dtypes: datetime64[us](1), float64(2), int64(2), str(9)
-- memory usage: 131.4 KB
```

Activate Windows

Step 3: Missing Value Detection

Null values were identified using Pandas functions and analyzed column-wise.

```
#Identifying Null Values
print("Missing / Null Values in each column:")
df.isnull().sum()

[ ]
```

```
-- Missing / Null Values in each column:
--
--      0
-- OrderID  0
-- Date      0
-- CustomerID  0
-- Product   0
-- Quantity  0
-- UnitPrice  0
-- ShippingAddress  0
-- PaymentMethod  0
-- OrderStatus  0
-- TrackingNumber  0
-- ItemsInCart  0
-- CouponCode  309
-- ReferralSource  0
-- TotalPrice  0
--
-- dtype: int64
```

Activate Windows

```
#CouponCode has null values
print("Records with null values in CouponCode column:S")
df[df["CouponCode"].isnull()]

[ ]
```

```
-- Records with null values in CouponCode column:S
--
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
8	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	NaN	Email	268.56
15	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	NaN	Google	473.96
17	ORD200017	2024-03-02	C60672	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	NaN	Referral	423.40
21	ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	NaN	Google	1371.80
26	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	NaN	Referral	297.75
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1172	ORD201172	2025-02-04	C28828	Desk	3	131.96	745 Main St	Credit Card	Pending	TRK32550359	8	NaN	Instagram	395.88
1184	ORD201184	2023-09-27	C10222	Laptop	1	204.08	979 Main St	Gift Card	Returned	TRK86297895	4	NaN	Google	204.08
1189	ORD201189	2024-08-17	C96136	Tablet	5	389.22	149 Main St	Cash	Cancelled	TRK60856682	7	NaN	Referral	1946.10
1192	ORD201192	2024-02-28	C72007	Chair	3	432.85	999 Main St	Online	Pending	TRK16240916	4	NaN	Instagram	1298.55
1196	ORD201196	2024-03-04	C20095	Monitor	2	662.53	778 Main St	Online	Cancelled	TRK69207593	5	NaN	Facebook	1325.06

309 rows × 14 columns

Step 4: Duplicate Record Detection

Duplicate records were identified and removed to improve data consistency.

```
> ~
#Identifying Duplicate rows
print("\nNumber of Duplicate Rows:", df.duplicated().sum())

[ ] Python

...

Number of Duplicate Rows: 0

#No duplicate records exist

[ ] Python
```

Step 5: Data Type Validation

Columns were checked and converted into appropriate data types where necessary.

```
> ~
print("\nCurrent Data Types:")
df.dtypes

[ ] Python

...

Current Data Types:

...
0
OrderID      object
Date      datetime64[ns]
CustomerID   object
Product      object
Quantity     int64
UnitPrice   float64
ShippingAddress object
PaymentMethod object
OrderStatus  object
TrackingNumber object
ItemsInCart  int64
CouponCode   object
ReferralSource object
TotalPrice   float64

dtypes: object

Empty markdown cell, double-click or press enter to edit.

#All columns are already in the correct data formats. No conversion is needed

[ ] Python
```

Step 6: Data Cleaning

Data inconsistencies were corrected and data formatting was standardized.

```
> ~
#Handling Missing Values
df['CouponCode']=df['CouponCode'].fillna('No Coupon Code')
df.isnull().sum()

✓ 0.0s Python

OrderID      0
Date          0
CustomerID    0
Product       0
Quantity      0
UnitPrice     0
ShippingAddress 0
PaymentMethod 0
OrderStatus   0
TrackingNumber 0
ItemsInCart   0
CouponCode    0
ReferralSource 0
TotalPrice    0
dtype: int64
```

Step 7: Verification

The cleaned dataset was verified to ensure accuracy and consistency.

7. Data Cleaning Tasks Performed

Missing Value Analysis

- Identified missing values in the dataset.
- Analyzed the impact of missing data.
- Applied suitable handling techniques.

Duplicate Record Removal

- Checked for duplicate records.

Data Type Check

- Examined data for inconsistencies.
- Ensured columns had appropriate data types.

Data Validation

- Verified the accuracy and completeness of the cleaned dataset.

8. Results

After performing data cleaning operations:

- Missing values were identified and handled.
- Duplicate records were removed.
- Data types were standardized.
- Dataset quality was significantly improved.
- Dataset became suitable for further analysis and visualization.

9. Key Observations

- Data quality improved after cleaning.
- No duplicate records were identified in the dataset.
- Missing value issues were addressed.
- Dataset consistency increased.
- The cleaned dataset became more reliable for analytics.

10. Challenges Faced

- Identifying suitable techniques for handling missing values.
- Detecting data inconsistencies.

- Maintaining data integrity while cleaning.
- Ensuring accuracy after preprocessing.

11. Conclusion

The Data Cleaning and Preprocessing project successfully improved the quality of the "Dataset for Data Analytics". Missing values, duplicate records, and inconsistencies were identified and handled appropriately. The dataset was also checked for duplicate records and no duplicates were found. The final cleaned dataset is accurate, reliable, and ready for further analysis, visualization, and machine learning applications.

13. References

1. Python Documentation
2. Pandas Documentation
3. Data Analytics Course Materials
4. Jupyter Notebook Documentation
5. DecodeLabs Industrial Training Kit